

The JiMS Chord-Naming System

Chords named in moveable-Do tonic solfa with a La-based minor. No pitch-names, anywhere.

0. The axiom: La-based minor

One choice underwrites everything: **moveable Do with a La-based minor**. The seven syllables

Do · Re · Mi · Fa · So · La · Ti

are welded to a single fixed diatonic collection. They *are* the scale degrees. Change the mode and the syllables do not move — only the tonic, marked on the JiMS staff by a diamond-shaped scale-dot, moves among them. La-mode's tonic is La; So-mode's is So.

This is what makes chord names possible at all. Because "So" always names the same note in the same collection, the stack of thirds above So is fixed once and forever — one shape, in every mode. In a Do-based minor, where the minor tonic is re-lettered Do and the syllables shift with each form of minor, "So7" names nothing definite until you also say *which* minor. La-based minor removes the ambiguity. The entire system rests on it.

1. The rules

1.0 Three registers, one hierarchy

Harmony contains three genuinely different kinds of chord, and JiMS names each in its own register. The registers are **not co-equal**, and the notation is honest about that:

1. **Tertian — derived from the scale**. The seven canonical third-stacks *fall out of* the diatonic scale and its modes. You do not specify them; you point at structure the scale already built. Named **XxN**. Clean **because** derived, not invented.
2. **Symmetric — off the scale, uniform**. The equal divisions of the octave — diminished seventh, augmented triad — arise from *no* diatonic mode. With nothing pre-built to point at, you describe them by a repeated generator: **g^N**.
3. **Root-relative — off the scale, ad hoc**. Chords like the augmented sixth are neither diatonic stacks nor uniform symmetric ones. You describe them by an explicit list of intervals above the root: **Fa:M3A6**.

Kludginess **increases with distance from the scale** — and that is information, not noise. Register 1 is the scale's own output; registers 2 and 3 are sequences imposed on it. A uniform notation would lie about how special the diatonic case is. There is no ideal single form, because the objects are not uniform.

1.1 The three forms

tertian: [Yy :] Xx N [+ Y, Z, ...] [/ Zz]

symmetric: g^N [/ Xx]

root-relative: Yy : {interval-list} [+ ...] [/ Zz]

Every operand is a solfa syllable — diatonic or extended-chromatic (Ra, Fi, Le, Te, Di, Me, Si, Ri, Li, ...) — **never a pitch-name**. Intervals in a root-relative list (M3, A6, ...) are named directly.

1.2 Register 1 — tertian chords

1. **Xx** — a diatonic syllable [Do Re Mi Fa So La Ti], naming that degree's **canonical stack**: the thirds obtained by climbing the diatonic scale from Xx. Xx is the chord's **identity**.
2. **N** — an odd numeral, 3...13, **truncating** the stack at the Nth degree. N carries all quality, because the canonical stack already encodes each third's species. Note count: (N+1)/2. **3** dyad · **5** triad · **7** seventh · **9** ninth · **11** eleventh · **13** full diatonic saturation.
3. **+ Y, Z, ...** — optional **adjunction tail**: further stack-degrees above N, strictly ascending, gaps preserved. Each +degree adjoins only itself. The base may be the bare root: **Do+5** is the power chord.
4. **Yy :** — optional **root**. Present $\Rightarrow$ **transplant**: build Xx's shape on Yy, computing upper syllables by interval; chromatic syllables fall out of the arithmetic. Absent $\Rightarrow$ native ($Yy = Xx$; one never writes $Xx:Xx$). **The colon flags a borrowed chord**.
5. **/ Zz** — optional **inversion**, defined as **bass-tone identity**: Zz is the lowest-sounding note. Absent $\Rightarrow$ root in bass. Zz in the chord $\Rightarrow$ inversion; Zz outside it $\Rightarrow$ pedal/foreign bass. One slash; membership decides.
6. **Precedence**: colon binds tighter than slash — assemble the chord, then set its bass.

1.3 Register 2 — symmetric chords

7. A symmetric chord has **no home degree** — no diatonic Xx generates it — so it is named by **generator and count**: $m3^N$ (the diminished family), $M3^N$ (the augmented family). N counts intervals, not scale-degrees.
8. g^N alone names the pure, rootless, placeless object.
9. g^N/Xx commits it to a position: the stack builds upward from Xx, with Xx in the bass. A symmetric stack has no competing root, so bass and generative origin coincide — the slash carries both. No **Yy :** slot is needed.
10. **Rotations are choices of /Xx** — re-rootings on the chord's own members. For symmetric chords, inversion and rotation are the same operation.

1.4 Register 3 — root-relative interval sets

11. A chord that is neither a diatonic stack nor a uniform symmetric one is named by an **explicit interval list above the root**: $Yy:\{list\}$. The intervals are reckoned **from the root**, not consecutively.
12. The paradigm is the augmented-sixth family. Core: $Fa:M3A6 = FaLaRi$, the Italian sixth. Members add root-relative intervals via the ascending **+** tail: French $Fa:M3A6+A4$, German $Fa:M3A6+P5$, Hendrix $Fa:M3A6+P5+m7$. See §7.
13. **/Zz** sets the bass as everywhere; the interval-list name itself is inversion-independent.

1.5 Scope

These rules name **every chord of tonal harmony that is a rooted stack of diatonic thirds (native or transplanted), plus the symmetric equal-division families, plus the root-relative interval sets**. That is the whole vertical vocabulary of Common Practice, blues, rock, and jazz.

Two boundaries remain — and both are the ceiling of *any* vertical chord-symbol, not JiMS gaps:

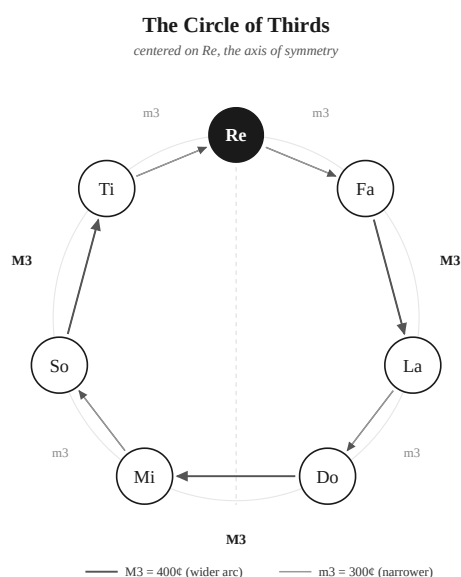
- **Linear sonorities** — cadential and passing six-fours, suspension and pedal chords, common-tone diminished sevenths — can be spelled but not meaningfully captured, because their identity is horizontal (voice-leading), not vertical.

- **Voicing-as-identity** — chords whose identity lives in register, spacing, timbre, or spectral fusion (a rootless quartal voicing; a #9 held as a self-standing tonic) — are named only to their skeleton, because voicing is deliberately unnotated.

The augmented sixth is **not** in this residue: register 3 names it fully (§7). Only its *resolution* — a voice-leading event — falls under the linear ceiling, where every chord's resolution falls.

2. The canonical stacks

Each diatonic degree has one fixed stack of thirds. All seven are rotations of a single circular pattern, because stacking thirds cycles **Do** → **Mi** → **So** → **Ti** → **Re** → **Fa** → **La** → **Do**. There is really one diatonic object — the third-cycle — and seven places to enter it.



The circle makes the claim visible: the seven notes sit on one loop of stacked thirds — three major (Do → Mi, Fa → La, So → Ti) and four minor — and each mode is simply a choice of *where to start reading*. The arcs are drawn to scale: a major third spans a wider arc than a minor third, and the seven together close exactly two octaves ($3 \times M3 + 4 \times m3 = 2400 \text{ ¢}$), returning to the start.

The loop is drawn **centered on Re**, which is its axis of reflection: the interval pattern reads the same clockwise and counter-clockwise from Re (m3, M3, m3, M3, m3, M3, m3), so the figure is mirror-symmetric about the vertical. This is why Re-mode (Dorian) is the one palindromic mode, and why Re is the natural center of the JiMS button-field. **Xx13**, the full stack from any degree, is one complete trip around this circle beginning at Xx; every shorter chord is a prefix of that trip.

Xx	3rd	5th	7th	9th	11th	13th
Do	M3	m3	M3	m3	m3	M3
Re	m3	M3	m3	m3	M3	m3
Mi	m3	M3	m3	M3	m3	m3
Fa	M3	m3	M3	m3	M3	m3
So	M3	m3	m3	M3	m3	m3
La	m3	M3	m3	m3	M3	m3
Ti	m3	m3	M3	m3	m3	M3

To read a chord: take Xx's row, keep the first $(N-1)/2$ thirds. **Do5** = M3-m3, the major triad. **So7** = M3-m3-m3, the dominant seventh — the only dominant-quality stack in the deck. **Ti5** = m3-m3, the diminished triad. Nothing is chosen; everything is revealed by how far N reaches.

The species are *generic* diatonic thirds — the larger (M3) and smaller (m3) thirds of the temperament. The pattern of larger and smaller around the cycle is invariant across the entire syntonic continuum; only the cent-sizes slide with the tuning. The canonical stacks are canonical across the temperament, not merely in 12-TET.

3. Worked examples

Name	Notes	What it is
Do5	Do-Mi-So	major triad (native)
So7	So-Ti-Re-Fa	dominant seventh (native)
Ti7	Ti-Re-Fa-La	half-diminished seventh (native)
Re13	Re-Fa-La-Do-Mi-So-Ti	full diatonic 13th on Re (Dorian)
Do+5	Do-So	power chord — root and fifth, no third, no species
Do5+9	Do-Mi-So-Re	add9 — triad plus ninth, no seventh
So7+9, 13	So-Ti-Re-Fa-La-Mi	dominant 13th, 11 omitted (the jazz voicing)
Re : So7	Re-Fi-La-Do	V7/V — the dominant shape transplanted onto Re
La : Do5	La-Di-Mi	Picardy tonic — the major triad transplanted onto La
Ra : Do5/Fa	Fa-La-Ra	Neapolitan sixth
Do5/So	So-Do-Mi	cadential six-four
m3 ³ /Ti	Ti-Re-Fa-Le	diminished seventh on Ti
M3 ² /Do	Do-Mi-Si	augmented triad on Do

4. The names carry function

Because the syllables *are* scale degrees, a JiMS name doubles as a function label — read against the tonic diamond on the staff:

- **So7** , diamond on Do: the chord on the fifth degree. **V**.
- **Re:So7** : the tonicizing shape (**So7**), rooted on the second degree. **V of V** — the colon says *borrowed*, the right side says *dominant shape*, the left side says *on Re*. The tonicization logic is on the face of the name.

Function is a two-place relation: **chord and mode** — not chord and key. The name supplies the chord relative to the tonic; the diamond's position among the scale-dots supplies the mode; together they fix the function. And the diamond distinguishes what Roman-numeral analysis must argue in prose: **tonicization** (diamond stays; **Re:So7** is V7/V) versus **modulation** (diamond moves; the same notes become the plain dominant of a new tonic). The distinction is a glyph position.

5. The symmetric families

5.1 Why they stand apart

The tertian grammar names a chord by its home degree. The symmetric chords — the equal divisions of the octave — have **no unique root**: any member can be the bottom. That rootlessness is precisely what makes them **pivot chords** (maximal voice-leading ambiguity, in Tymoczko's geometry the center of chord space), and precisely why they fall outside the **Yy:Xx** grid. JiMS refuses to assign them a home degree because they do not have one. The property that excludes them from register 1 and the property that makes them pivot are the same property.

5.2 The minor-third stack

- **m3²** — diminished triad (3 notes)
- **m3³** — diminished seventh (4 notes)
- **m3⁴** — octave closure (period 4)

m3³/Ti = Ti-Re-Fa-Le, the leading-tone diminished seventh. Its rotations — **m3³/Ti** , **/Re** , **/Fa** , **/Le** — are four spelled fingerings that coincide in sound only at $g = 700$ and are audibly distinct everywhere else on the continuum. **Modulation-by-pivot is modulation-by-respelling**: change the **/Xx** , and the change of name is the reinterpretation.

5.3 The major-third stack

- **M3²** — augmented triad (3 notes)
- **M3³** — octave closure (period 3)

M3²/Do = Do-Mi-Si. Rotations **/Do** , **/Mi** , **/Si** — three fingerings, coincident only at $g = 700$.

5.4 Side by side

	minor-third stack	major-third stack
generator	m3	M3
triad	m3 ² diminished	M3 ² augmented
tetrad	m3 ³ diminished seventh	—
octave closes at	m3 ⁴ (period 4)	M3 ³ (period 3)
rotations	4	3
primary use	pivot / linear connector	pivot / linear connector

5.5 The symmetry is a tuning artifact

The perfect symmetry that lets these chords pivot freely — that makes their rotations indistinguishable — exists **only at g = 700**, where the octave divides exactly. Off 700, the divisions are unequal, the rotations sound different, and each /Xx spelling is a determinate chord. A rank-2 JiMS button-field represents this **without wolves**: m3 is one fixed button-shape everywhere, M3 another, and the notation never collapses the spellings. The free pivot is a property of the 12-TET *readout* — which JiMS lets you see through rather than be trapped by.

6. Grammar summary

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chord      := tertian | symmetric | rootrel      ; the three registers

; register 1 – derived from the scale (privileged, clean)
tertian    := [ root ":" ] degree numeral [ tail ] [ "/" bass ]

; register 2 – off the scale, uniform (repeated generator)
symmetric  := generator "^" count [ "/" bass ]

; register 3 – off the scale, ad hoc (explicit intervals from root)
rootrel    := root ":" interval { interval } [ tail ] [ "/" bass ]

root       := syllable          ; any solfa syllable
degree     := "Do"|"Re"|"Mi"|"Fa"|"So"|"La"|"Ti"    ; home degree (identity)
numeral    := "3"|"5"|"7"|"9"|"11"|"13"            ; truncation, odd
tail       := "+" adjunct { "," adjunct }           ; ascending, gaps kept
adjunct    := syllable-degree | interval            ; note above the root
generator  := "m3" | "M3"
count      := integer                      ; number of stacked intervals
interval   := "M3" | "A6" | "A4" | "P5" | "m7" | ... ; interval above the root
bass       := syllable                      ; lowest-sounding note

; colon binds tighter than slash
; register 1 reads intervals consecutively; register 3 reads them from the root
; absent root ⇒ native (root = degree); colon present ⇒ transplant / root-relative
; absent bass ⇒ root in bass
; syllables are diatonic or extended-chromatic; never pitch-names

; the registers are NOT co-equal: kludginess increases with distance from
; the scale. Register 1 is derived; registers 2 and 3 are described. That
; the notation looks different for them is honesty, not defect.
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7. The augmented sixth — and the Hendrix chord

The augmented sixth long looked like the residue: its defining interval is an augmented sixth, not a stacked third. In JiMS it needs no special apparatus. The family has one canonical core, its members are that core plus ordinary + additions — and the same construction yields the **Hendrix chord (7#9)**. They are one graded object.

Fix the mode: **La-mode** — tonic La, dominant Mi, third-degree Do anchoring the fifth-chain. Every augmented sixth is spelled from **Fa**, the sixth degree (§7.4 says why).

7.1 The core: **Fa:M3A6**

The augmented sixth is two intervals above one root:

$$\mathbf{Fa:M3A6} = \mathbf{Fa} + (\mathbf{M3} \rightarrow \mathbf{La}) + (\mathbf{A6} \rightarrow \mathbf{Ri}) = \mathbf{FaLaRi}$$

This is the **Italian sixth** — the minimal augmented sixth. Remove any note and the chord is gone. The name **M3A6** is the family identifier: the major third and the defining augmented sixth, both from the root, in the same root-relative frame as the **+** additions.

And **Ri** is chain position **+9** — the augmented second above Do, the tonic's **septimal blue third** (7/6 in the meantone / 31-TET region; "the blue 3rd" of the Blues Continuum). It is a nameable syllable, not a chromatic accident.

7.2 The members

Each member is the core plus root-relative additions, listed ascending:

chord	name	notes	adds
Italian (core)	Fa:M3A6	FaLaRi	—
French	Fa:M3A6+A4	FaLaTiRi	Ti (A4)
German	Fa:M3A6+P5	FaLaDoRi	Do (P5)
Hendrix	Fa:M3A6+P5+m7	FaLaDoMeRi	Do (P5), Me (m7)

The Italian is the bare skeleton: major third plus blue third. The French adds the supertonic, whose two tritones — diatonic Fa ↔ Ti, chromatic La ↔ Ri — give the whole-tone tint. The German adds the fifth, completing the diatonic Fa triad (Fa5+Ri). The Hendrix chord adds fifth and seventh both, completing the dominant seventh (Fa7+Ri).

The German sixth is the Hendrix chord minus its b7.

Two naming registers coexist. The generative name (Fa:M3A6+P5) is a recipe — core, then additions by ascending interval. The enumerative name (FaLaDoRi) lists the notes in pitch order. Both denote the same chord; the first carries family identity, the second is the flat spelling. Inversion, as everywhere, is /Zz : Fa:M3A6+P5/Do is the German sixth over Do. The name itself never changes with inversion.

7.3 The blues identity

The augmented sixth's manufactured note and the blues third are **the same note** — Ri, chain +9. Only a fixed-syllable notation makes this visible. The interval Fa ↔ Ri is therefore a **septimal augmented sixth**: in the meantone region, where 225/224 vanishes and the augmented sixth lands near 7/4, it approximates a just septimal interval and **rings** — exactly as the blues notes ring, and audibly so on the meantone-tuned, seventh-partial-rich keyboards the chord was born on.

The Common Practice augmented sixth is, structurally, the era's blues chord.

7.4 Why it exists: a prosthesis on a prosthesis

The chord does one thing: drive by contrary semitone onto the dominant — Fa → Mi from above, Ri → Mi from below. It is spelled from Fa because in minor the sixth degree sits a semitone above the dominant; that semitone is the down-resolving voice. In the bright modes the sixth degree sits a whole tone above the dominant, so there the augmented sixth must *borrow* a lowered sixth from the parallel minor — a mixture chord in major, native only to minor.

The deeper rationale is polemical, and it completes the arc that runs through the raised leading tone and the Picardy third. Common-practice minor covets the one cadence Ionian owns natively: the leading-tone V7 → I. Minor lacks it, so the tradition manufactures a dominant by raising the seventh (So → Si). The augmented sixth is a **second fabrication bolted onto the first** — a chord present in no diatonic mode, conjured from whole cloth, whose sole

purpose is to hurl itself by double semitone onto that already-manufactured dominant. A cadential prosthesis on a cadential prosthesis: the era's most naked admission of Ionian-cadence desperation, fabricating chords from nothing so that minor can counterfeit the cadence Ionian gets for free.

The septimal blue third is the **incidental residue** of the counterfeiting. The tradition reached for "a leading tone to the dominant," and a leading tone to the dominant is exactly where a 7-limit note falls. The Common Practice minted the blue note by accident, dressed it as a chromatic alteration, and let it ring on meantone keyboards for two centuries before anyone recognized what it was.

7.5 The enharmonic pivot

At $g = 700$ only, $Ri \equiv Me$. There the German sixth collapses to $Fa7$, a plain dominant seventh — the classic remote-modulation pivot, executed by the single respelling **Ri** $\leftrightarrow$ **Me**, which is the modulation. The same collapse folds the Hendrix chord into a dominant seventh in which Me serves as both $b7$ and $\sharp 9$ — which is exactly why the Hendrix chord sounds like it holds a major third and a minor third at once. **The German sixth is the Hendrix chord at the moment its two blue notes fuse into one.**

Off $g = 700$, Ri (chain +9) and Me (chain -3) are different pitches. The pun dissolves; neither chord is a dominant seventh. On a rank-2 button-field they were always different buttons — the pivot is a property of the 12-TET readout, which JiMS lets the composer exploit or refuse, rather than have the instrument impose.

7.6 The ledger

- **Pitch content: addressed, with no new machinery.** Every augmented sixth, and the Hendrix chord, is $Fa:M3A6$ plus ascending additions.
- **Function: the one general limit.** The characteristic outward resolution is a voice-leading event, and JiMS names vertical objects, not resolutions. That is the ceiling every chord-symbol meets; the augmented sixth just meets it conspicuously.

8. Why this matters

Understand the diatonic scale, its seven modes, and the third-stack each mode generates — plus the two symmetric stacks and the one root-relative family — and you understand the **harmonic vocabulary** of Common Practice, blues, rock, and jazz as a single generative system. Not a catalogue to memorize: a small set of generators to compute from.

The economy is measurable. A chord-book is a printed-out *product* — root $\times$ key $\times$ quality $\times$ voicing, hundreds to thousands of entries. JiMS stores the *factors*: the root is one syllable; neither mode nor key appears in the name at all (the mode is the staff's diamond — which scale-dot is tonic; the key is the pitch-name or frequency written beside it); the quality is a canonical stack or a short recipe; the inversion is one slash. **A multiplication table becomes multiplication.**

Jazz's alteration labels are Ionian-relative relabeling. Jazz's own chord-scale theory concedes that a chord is its parent mode's third-stack — a minor 13th is the Dorian mode stacked in thirds. But jazz's *symbols* measure every $b9$, $\sharp 11$, $b13$ against the major scale built on the chord's root: a chord-local Ionian ruler. JiMS names the parent-mode stack directly — $Re13$, $So13$, $Ti13$, each unique — and so needs none of the $b/\sharp$ apparatus, which existed specifically and only to express relationship to the Ionian stack.

What remains of "understanding jazz" is not more chords. It is the horizontal layers no vertical notation supplies: voice-leading, functional succession, and the voicing/register/timbre dimension where sound outruns spelling. JiMS collapses the vertical, tonally-functioning universe to a small generative core — completely — and stops exactly where music stops being vertical pitch-content.

8.1 The diatonic is the first instance, not the foundation

The scale-dots on the JiMS staff declare which scale is in force, and everything above recomputes from that declaration. The diatonic seven is one setting. Other tonally-useful scales — well-formed ones, and non-well-formed ones like the double harmonic — are peers under the identical dot-and-stack rule; they simply generate different canonical stacks.

One striking consequence: the rootless orphans of the diatonic frame — the augmented triad, the diminished seventh, named $M3^N$ and $m3^N$ because no diatonic degree generates them — become **native XxN chords** in a scale whose stacks *do* generate them. Rootlessness was never absolute; it was relative to which scale the dots declared. The real primitive of the system is not the diatonic scale. It is **a dotted, tonally-useful scale and its stack of thirds** — of which the diatonic is the first and finest instance.

Four orthogonal axes: identity (the stack or recipe), placement ($Yy:$), inversion ($/Zz$), voicing (deliberately unnotated, left to the performer).